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Development of a multi-facet facial recognition model for institutional attendance activities during examination scenario

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Abstract

Attendance recording and verification are essential in educational institutions, especially during examinations, where ensuring each student's presence is important. Conventional attendance systems rely on manual methods such as name calls or signature verification, which are often time-consuming and prone to errors. Although various facial recognition systems have been introduced to address these challenges, many state-of-the-art models still struggle to accurately identify students under varying conditions, including differences in lighting, head orientations, and facial appearances influenced by accessories. To overcome these limitations, this study proposes a multi-facet facial recognition model based on the K-Nearest Neighbors (KNN) algorithm for recording and verifying student attendance during examination scenarios. The model was trained to operate reliably under different lighting, head orientations, and use of accessories, specifically glasses and beards, by leveraging pre-registered students' image data captured across multiple settings, forming a training dataset of 990 images. The process involved face detection using a pre-trained Haar Cascade classifier, followed by face alignment and encoding into 128-dimensional vectors using a ResNet-based approach, which were then classified by the KNN algorithm. Evaluation across static images, video, and real-time camera feeds achieved an overall accuracy of 99.00% with a response time of 6.00s seconds. Compared to traditional manual attendance systems, the proposed model provides a solution for examination verification, effectively mitigating recognition challenges caused by varying environmental and facial conditions.

Keywords Computer vision, Examination attendance authorization, Facial recognition, Multi-facet recognition

1 Introduction

An examination serves as a fundamental method for assessing, measuring, and validating an individual's knowledge, understanding, and competence within a specific subject or field [1]. For an examination to be deemed credible, it must follow certain fundamental guidelines: it must be administered impartially, without bias, cheating, or malpractice



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of any kind [2–4]. One of the most prevalent types of examination malpractice is candidate impersonation, in which a more experienced person is hired to take the assessment in a candidate's place [5, 6]. During the examination, invigilators use identification (ID) cards to confirm the candidates' identities, usually containing the candidate's name, photo, and other pertinent information [7]. The invigilator then verifies the candidate's identity by comparing their face to an ID card photo. Unfortunately, there is a serious problem with this approach: the ID card's photo can be that of an imposter, even while the rest of the information matches that of the real candidate. This flaw compromises the integrity of the assessment process by exposing a serious weakness [8, 9].

Due to this, maintaining the integrity of academic examinations is increasingly challenged by candidate impersonation, where manual verification using identification cards remains vulnerable to substituted photographs. While earlier automated attendance systems utilizing clickers, smart cards, RFID, and fingerprint scanning sought to mitigate these human errors, they remain susceptible to proxy usage or require physical contact. Since biometric information is specific to each person and can reliably differentiate one from another, it provides a very efficient means for verification. Hence, facial recognition has emerged as a highly dependable, contactless alternative for verifying an individual's actual presence and securing the assessment process against impersonation [10].

According to Chin [11], facial recognition is a biometric method that uses a comparison between an image taken in real-time and previously recorded images in a database to identify people. Due to its authenticity verification capabilities, user-friendly nature, and contactless operation, facial recognition stands out from other biometric techniques like fingerprints, making it an excellent option for applications where accuracy, efficiency, and security are important [12–14].

However, deploying these systems in examination environments introduces significant challenges. Current approaches often struggle to accurately identify students under varying real-world conditions, experiencing severe performance degradation when simultaneously confronted with uneven lighting, non-frontal head orientations, or facial appearances influenced by accessories like glasses and beards.

1.1 Motivation, aim, and scope

The primary motivation of this research stems from a critical gap in current examination security systems. While conventional manual verification remains vulnerable to candidate impersonation, state-of-the-art deep learning attendance solutions often demand high computational overhead and suffer severe performance degradation when confronted with simultaneous real-world disruption such as uneven classroom lighting combined with angular head poses and facial accessories. Prolonged processing times in these heavy architectures creates unacceptable bottlenecks for student ingress. Therefore, there is a pressing need for a system that balances robust occlusion handling with low latency processing.

The aim of this study is to develop and evaluate a lightweight, multi-facet facial recognition framework optimized specifically for dynamic institutional examination scenarios. To achieve this, the specific objectives are to:

- i. Develop a recognition pipeline that integrates Haar Cascade detection, ResNet-based feature encoding, and a K-Nearest Neighbors (KNN) classifier to ensure low-latency identity verification.

- ii. Evaluate the model's robustness against three concurrent sources of variance common in examination environments specifically illumination changes, non-frontal head orientations, and physical occlusions.
- iii. Compare the proposed framework against alternative lightweight models to validate its computational efficiency and real-time deployment suitability.

The scope of this research is focused strictly on access control and identity verification within educational examination settings. The experimental evaluation is bounded to recognizing registered individuals under predefined environmental and physical variations to prove the viability of the multi-facet approach.

1.2 Original contribution and novelty

To novelty of this study lies in its hybrid multi-facet pipeline. Unlike recent models that optimize for isolated variables like only mask detection or only illumination using heavy Convolutional Neural Networks (CNNs), the proposed method utilizes ResNet strictly to map complex, occluded facial landmarks into a 128 dimensional Euclidean space. By subsequently passing these embeddings to a lightweight, distance weighted KNN classifier, the model resolves concurrent variations specifically lighting, pose and accessories together without the high computational latency typical of end-to-end deep learning pipelines.

The remainder of this manuscript is organized as follows: Sect. 2 reviews related works and justifies the proposed method against recent literature. Section 3 details the implementation, including dataset splits and architectural parameters. Section 4 presents the experimental analysis, including comparative baselines. Finally, Sect. 5 discusses the conclusions and limitations.

2 Related works

The integration technology into education has sparked increased interest in computerized platforms for classroom examination identity verification. While early attendance systems relied on localized databases and basic feature extraction, recent technological advancements have enabled the development of robust, AI-powered solutions. The current literature discusses a high level of development within biometric automated attendance; however, achieving seamless, real-time integration under dynamic classroom conditions remains an active area of research.

2.1 Traditional and deep learning approaches to automated attendance

Early computerized attendance systems predominantly relied on handcrafted feature extraction and traditional machine learning classifiers to automate student verification. Foundational frameworks frequently utilized Viola-Jones technique for face detection paired with Local Binary Pattern Histograms (LBPH) to encode facial characteristics [15–17]. These studies demonstrated that standardizing facial cropping and local texture patterns could successfully replace manual roll calls, significantly reducing human error and administrative time by logging data directly into SQL databases. However, these traditional methods require strictly facial alignment and are highly sensitive to complex classroom environments.

To overcome the representational limits of traditional models, recent research has shifted towards deep learning architectures and cloud-based infrastructure. Comparative

analysis by Pattnaik and Mohanty [18] and Prakash et al., [19] highlighted the superior feature extraction capabilities of Artificial Intelligence APIs, such as Amazon Web Services, noting their enhanced accuracy when processing low-resolution images. Similarly, deep learning architectures have demonstrated strong performance in real-time facial analysis tasks beyond identity recognition. Muthukumar, [20] demonstrated that deep learning models can achieve high accuracy in real time facial sentiment recognition, confirming that convolutional neural network based approaches are capable of processing facial features rapidly under varying conditions. However, these architectures typically require substantial computational resources, which introduces latency challenges that are particularly problematic in high-throughput environments such as examination halls where multiple students must be verified within a constrained time window. Building on this need for real-time efficiency, state-of the art object detection algorithms have recently been adapted for continuous facial tracking. Notably Lateef and Kamil [21] and Febri Ananda et al., [22] Successfully deployed YOLOv7 algorithms to process live IP camera feeds in educational environments. By leveraging deep learning on extensive student datasets, these systems achieved verification accuracies exceeding 90%, establishing a new baseline for automated attendance and proving that neural networks can handle the rapid throughput required for institutional environments.

2.2 Facial recognition under occlusion and environmental variations

Addressing the specific physical occlusions, Kamil et al., [10] developed an online system combining facial recognition with face mask detection to enhance security and compliance. The system's effectiveness in practical applications was demonstrated by the results, which showed that it attained an accuracy of roughly 81.8% for facial recognition and 80% for face mask detection. However, while this approach was effective for isolated recognition and face mask detection, there is a need for broader occlusion robustness in unpredictable education settings. Expanding on this, Patel et al., [23] proposed an attendance framework integrated with dynamic preprocessing filters to resolve partial facial occlusions. Their evaluations specifically highlighted severe performance degradation when standard models are confronted with everyday accessories, such as eyeglasses, and dense facial hair, without specialized feature alignment.

Beyond occlusions, varying illumination and non-frontal head poses represent critical, overlapping hurdles in classroom examination environments. Afianto et al., [24] evaluated automated attendance systems under fluctuating lighting, concluding that uneven illumination drastically disrupts traditional feature extraction. Concurrently, studies focusing on angular head orientations Matar, [25], have analyzed the fragility of facial embeddings when subjects are not looking directly at the camera. These evaluations systematically demonstrate that off-angle head orientations especially when combined with poor lighting or accessories largely lower similarity scores. This consensus in the literature underscores the necessity for context aware thresholding and robust encodings.

Despite these advancements, a prominent research gap remains computational latency. While heavy Convolutional Neural Networks (CNNs) and deep learning models can technically resolve isolated variations, they frequently demand significant computational resources. As highlighted by, deploying these resource-intensive architectures for real-time student verification often results in processing bottlenecks, creating unacceptable delays at examination hall entrances. Therefore, the current literature indicates the

pressing need for models that can balance robust feature extraction against compounding variations (lighting, pose, and accessories) with lightweight, accurate low latency classification.

2.3 The need for multi-facet recognition and low-latency processing in dynamic examination scenarios

Despite the effectiveness of existing facial recognition systems, significant research gaps persist concerning their robustness when subject conditions vary. Prior studies typically optimize for isolated variables, such as illumination or the presence of a mask, while overlooking the compounded effects of multiple concurrent disruptions [10, 24]. Specifically, many existing frameworks lack robust facial embeddings and context-aware feature alignment, causing their similarity scores to degrade when confronted with off angle head poses or accessories like glasses [23, 25]. In addition, while heavy deep learning architectures offer higher accuracy under these variations, they frequently introduce significant computational latency and hence, in high-stakes institutional scenarios, such as examination hall entry, prolonged processing times can create unacceptable bottlenecks for student ingress [26].

To address these limitations, this study introduces a lightweight, multi-facet facial recognition model. Here, the term “multi-facet” refers specifically to the model’s ability to handle three simultaneous sources of variance: lighting changes, different head-pose orientations, and facial accessories, specifically glasses and beards. In contrast to conventional deep learning pipelines, which often experience substantial performance degradation under combined variations or suffer from high latency, the proposed approach integrates ResNet-based feature extractor with a K-Nearest Neighbors (KNN) classifier. This approach maintains high-accuracy identity verification across all three factors concurrently, ensuring both reliable attendance recognition in examination settings (Table 1).

3 Methods

The process depicted in Fig. 1 begins with a camera module capturing a live video feed, where each frame undergoes preprocessing through OpenCV for grayscale conversion and histogram equalization to optimize detection conditions. Haar Cascade face detection then scans these images using its pre-trained classifier, identifying potential face regions through a sliding window approach that evaluates Haar-like features at multiple scales. Detected faces are cropped to their bounding box coordinates. Next, the cropped image undergoes an embedding process where face locations are initially identified using the Histogram of Oriented Gradients (HOG) with a linear support vector machine (SVM) algorithm to extract the unique 68 landmark features of the face before a ResNet model encodes these features into a one hundred and twenty-eight (128) dimensional embedding vector, which is normalized using NumPy for comparison. The multi-facet model is then trained based on the computed 128-dimensional embeddings (i.e., facial encodings) for determining a match if the similarity between the input and stored facial encoding exceeds a configured threshold. Recognized individuals are logged into the attendance database with a timestamp and identification (ID), while unrecognized faces trigger an *unknown person* protocol. OpenCV then displays the recognition result with

Table 1 Comparative summary of recent literature (2020–2025) and the proposed methodology

Reference	Methodology	Application strength	Limitations
Khuran et al., [16]; Trivedi et al., [17]; Bairagi et al., [15]	Viola-Jones, LBPH, and OpenCV-based frameworks.	Computationally inexpensive; highly effective for standardizing automated attendance and fast database logging using basic camera setups.	Highly sensitive to uneven lighting; requires strict facial alignment; struggles with physical occlusions like glasses or dense beards.
Pattanaik and Mohanty, [18]; Prakash et al., [19]	Cloud-based Artificial Intelligence (AI) APIs (e.g., AWS).	Superior feature extraction capabilities; maintains high accuracy even when processing low-resolution classroom images.	Relies heavily on continuous internet connectivity; cloud processing latency can create unacceptable bottlenecks for real-time student ingress.
Lateef and Kamil, [21]; Ananda et al., [22]	YOLOv7 Object detection (deep learning).	Excellent for continuous, multi-face tracking in live educational environments using standard IP cameras.	High computational resource demand; deploying these heavy architectures on edge devices causes processing delays.
Kamil et al., [10]	Support vector machine (SVM) with synthetic data.	Highly effective for identity verification under isolated physical occlusions (e.g., pandemic-era face masks).	Optimizes only for isolated variables; lacks evaluated robustness for compounding daily variations (e.g., glasses combined with poor lighting).
Matar [25]; Afiyanto et al., [24]	Advanced CNNs, facial embeddings, dynamic preprocessing.	Provides deep, comprehensive analysis of facial robustness against varying lighting intensities and facial angles.	Reveals that similarity scores systematically degrade when off-angle poses combine with accessories without specialized context-aware feature alignment.
Swaminathan et al., [27]; Sasikumar and Nagarajan [28]	Multi-factor authentication and sentiment recognition architectures.	Extends deep learning biometric security to broader domains, including cloud infrastructure and humanoid robotics.	Not optimized for the unique, concurrent disruptions (pose, lighting, accessories) required for high-throughput examination entry.
The proposed model	Haar Cascade + ResNet-128D + KNN	Maintains high-accuracy identity verification across concurrent disruptions (lighting, pose, accessories) with rapid, low-latency classification.	Faces occasional “Failure to Detect” in extreme acute angles due to Haar Classifier limits, requiring temporal video redundancy.

bounding annotations before the system loops to process the next frame, completing the continuous attendance cycle.

3.1 Data collection

A total of 990 facial images were collected from three students using the camera module. The distribution of images across the participants are as follows: Student 1 contributed 330 images, Student 2 contributed 330 images, and Student 3 contributed 330 images.

3.2 Data pre-processing

3.2.1 Image resizing

To ensure uniformity across the entire dataset and to meet the requirements of the facial recognition model, each cropped face image was rescaled to a fixed size of 400×400 pixel using OpenCV's *resize* function. The resizing process underwent two stages to resize the image. The first stage involved coordinate mapping. Bilinear interpolation was then used to compute the pixel values of the resized image by calculating the intensity of a pixel in the resized image as a weighted average of the four nearest pixels in the original image.

Standardizing dimensions not only guaranteed that every sample contributed the same number of pixels for feature extraction but also simplified memory management, accelerating both training and inference by reducing computational costs.

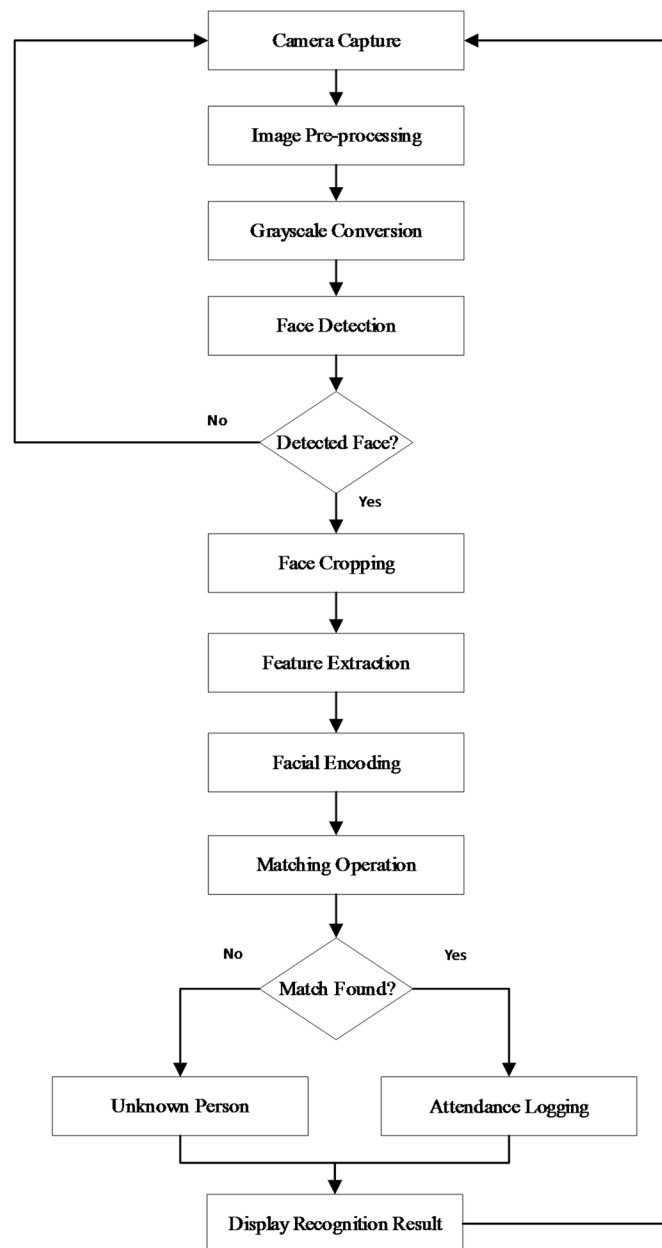


Fig. 1 The multi-facet facial recognition model system flow chart

3.2.2 Grayscale conversion

Following the image resizing step, the next pre-processing technique applied was gray scaling. This technique transformed the resized 400×400 -pixel color images from the BGR color space captured by OpenCV to grayscale, where each pixel's intensity was computed using the weighted sum procedure.

The conversion to grayscale was important since the facial recognition model primarily depended on structural and textural features rather than color, which was not a decisive factor in recognizing students. By reducing each image's data from three color channels to a single intensity channel, gray scaling significantly decreased the computational overhead, making it ideal for real-time processing.

3.3 Data preparation

To ensure good performance of the model across the different conditions, the data preparation process incorporated class balancing to mitigate bias and improve the generalization capability of the model. A total of 990 images were curated and used for the training process, distributed across various categories.

To also prevent data leakage and ensure unbiased evaluation, strict subject-wise and temporal separation was enforced between the training and testing datasets. The 990 images selected for training were curated from independent recording sessions distinct from the 3000 images reserved for testing. This ensured that the KNN classifier was evaluated solely on “unseen” data samples in order to reflect genuine feature generalization rather than overfitting to specific acquisition artifacts.

3.4 Haar Cascade frontal-face classifier for face detection

Since facial data was the primary focus, a pre-trained Haar Cascade face classifier was used for the initial facial detection process. While deep learning detectors offer higher accuracy, the Haar Cascade classifier was specifically chosen for its computational efficiency on low-resource hardware. The process of detecting frontal faces by the classifier consisted of a combination of four distinct techniques, namely: Haar-like features, integral image, adaptive boosting (Adaboost), and a cascade of classifiers. The detection capability of this classifier relied on the Haar-like features illustrated in Fig. 2. These rectangular features were specifically selected because they align with the natural shading properties of a human face. For instance, the feature detecting that the eye region is naturally darker than the upper cheeks. By relying on these structural contrasts rather than complex pixel-level patterns, the system effectively captured the face geometry required for the subsequent alignment stage.

Furthermore, to ensure the low latency required for real-time examination entry, the system employed the Cascade Structure depicted in Fig. 3. This architecture organized the detection process into multiple stages of increasing complexity. As shown in the figure, simple features were applied in the early stages to rapidly reject the vast majority

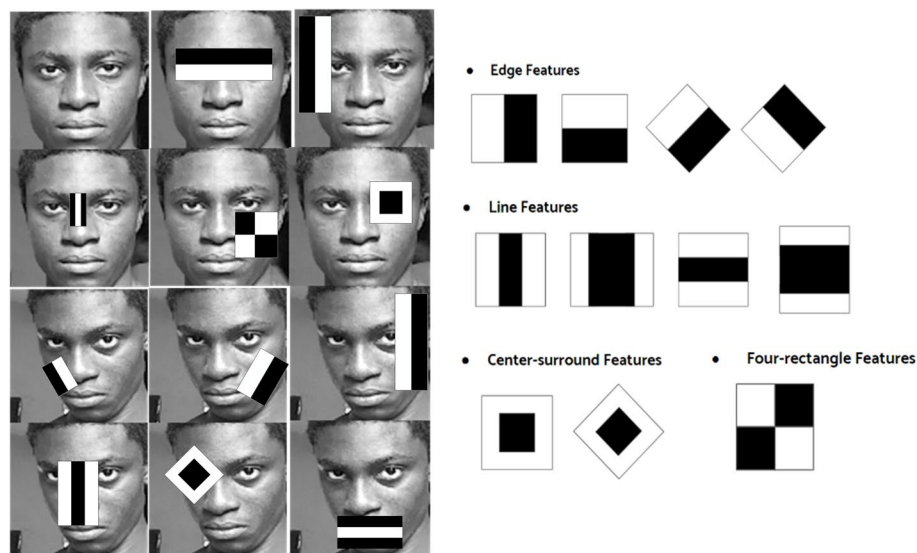


Fig. 2 The Haar-like features

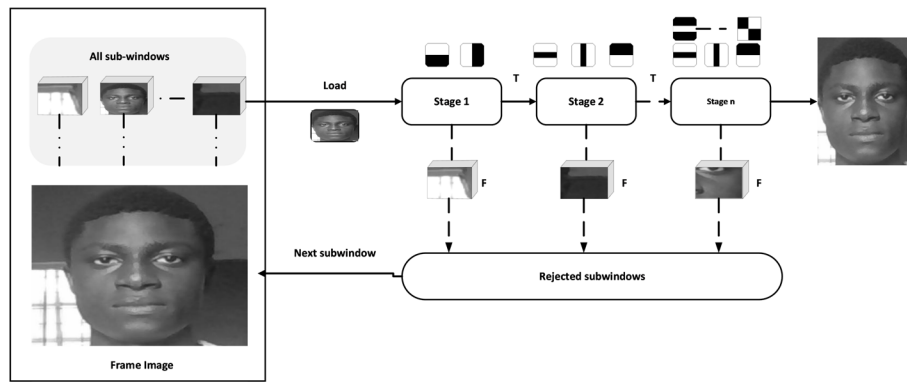


Fig. 3 The Cascade structure of the Haar-Cascade classifier

of non-face background regions (e.g. walls or door). This “early rejection” mechanism ensured that computational power was reserved strictly for promising regions, allowing the system to scan high-resolution video frames within milliseconds, an important choice for minimizing queues during student ingress.

3.5 Feature extraction

To capture student attendance irrespective of subtle changes in appearance, facial features were extracted using the Histogram of Oriented Gradients (HOG) algorithm. Unlike pixel-intensity matching, which is highly sensitive to ambient lighting, the HOG descriptor characterized local object appearance through the distribution of intensity gradients. This design choice was important for the examination scenario, as it allowed the system to align facial landmarks (e.g. eyes, jawline) based on structural shape rather than absolute brightness, providing a normalized input for the subsequent encoding stage.

However, since the input image had already been converted into a grayscale representation to simplify the gradient computation in the initial face detection process with the Haar Cascade frontal face classifier, gradients were then calculated for each pixel using the Sobel Operator, defined as:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} * I \quad (1)$$

$$G_y = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} * I \quad (2)$$

Where, G_x and G_y represents the horizontal and vertical gradient components, respectively, and I represents the image intensity.

Next, the gradient magnitude (G) and the orientation θ were then computed in Eq. 3 as:

$$G = \sqrt{G_x^2 + G_y^2}, \theta = \tan^{-1} \left(\frac{G_y}{G_x} \right) \quad (3)$$

These gradients were quantized into discrete orientation bins (typically 9 bins over 0° to 180°) within local cells (e.g., 8×8 pixels). The HOG descriptor was then formed by

normalizing these histograms across larger blocks (e.g., 2×2 pixels) to account for illumination changes, using the block normalization factor shown in Eq. 4:

$$V_{norm} = \frac{v}{\sqrt{\|v\|_2^2 + \epsilon}} \quad (4)$$

Where, v is the vector of histogram values, $\|v\|_2^2$ is the L2 norm, and ϵ is a small constant to avoid division by zero.

The resulting HOG features were fed into the pre-trained linear SVM classifier, which classified regions as faces or non-faces based on a sliding window approach across the image pyramid (i.e., upsampling to detect faces at multiple scales). The resulting output was now a set of bounding box coordinates (top, right, bottom, and left) for each detected face, which was returned as the face locations of that particular image.

3.6 Feature encoding

Once face locations were identified, the system utilized a ResNet architecture to generate a 128-dimensional embedding for each student. Preliminary tests utilizing raw pixel matching demonstrated poor accuracy due to high sensitivity to misalignment and accessories. Consequently, a deep learning based approach was adopted to abstract facial features into a compact Euclidean space.

The ResNet model mapped facial landmarks into a 128-dimensional vector space where the distance directly corresponded to facial similarity. This optimization was driven by the triplet loss function, which enforces that the distance between an anchor face (f_a) and a positive match (f_p) is minimized, while the distance to a negative match (f_n) is maximized by a margin α . This defined mathematically as:

$$L = \sum_i \max\left(0, \|f_a - f_p\|_2^2 - \|f_a - f_n\|_2^2 + \alpha\right) \quad (5)$$

Where, f_a is the anchor face encoding, f_p is the positive face encoding (i.e., the same person as the anchor), f_n is the negative face encoding (i.e., a different person from the anchor), $\|\cdot\|_2$ is the Euclidean distance, and α is the margin parameter, which is typically 0.2.

This encoding strategy ensured that the model produced separable representations for the three identities, effectively clustering a student wearing glasses and the same student without glasses as the same identity, a capability that strictly pixel-based comparison methods often lack (Fig. 4).

3.7 Model development

The K-Nearest Neighbors (KNN) algorithm was used as the core classifier. This was a non-parametric approach chosen over complex neural network classifiers to allow for incremental learning where new students could be added to the database simply by storing their embeddings without retraining the entire network. To classify a student, the Euclidean distance between the test encoding (x) and stored training encodings (x_i) was used and to improve robustness against outliers, a distance-weighted voting scheme was implemented. Unlike uniform voting, this method assigned higher importance to

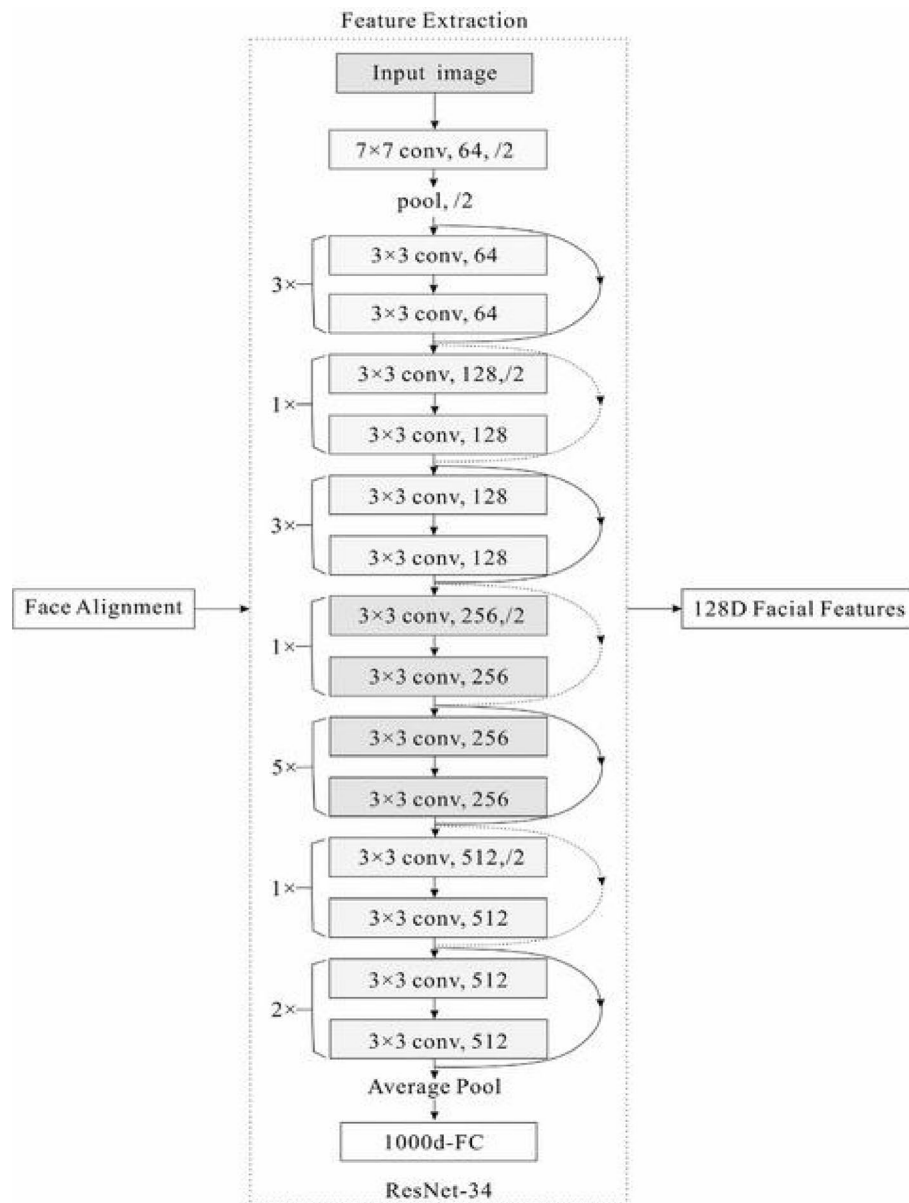


Fig. 4 The ResNet model architecture for the facial encoding process

the closest training samples, ensuring that a single very similar image outweighs multiple distinct neighbors.

To validate the choice of the KNN, a comparative benchmarking framework was done to evaluate the KNN model against two other standard lightweight classifiers: a support vector machine (SVM) and a shallow neural network (SNN). The SVM was configured with a linear kernel to align with the high-dimensional properties of the feature space, while the SNN was implemented as a Multi-Layer Perceptron (MLP) featuring a single hidden layer of 64 neurons and the Adam optimizer. The mathematical formulations for the KNN model are presented below.

3.8 Mathematical formulations for the multi-facet facial recognition model

3.8.1 Euclidean distance calculation

The Euclidean distance was used to measure the similarity between two 128-dimensional vectors, x the test encoding of a student's face and x_i the training encoding with smaller distances indicating greater similarity. It is mathematically given as:

$$d(x, x_i) = \sqrt{\sum_{j=1}^{128} (x_j - x_{i,j})^2} \quad (6)$$

Where, x_j and $x_{i,j}$ are the j^{th} components of the test and training encoding vectors, respectively, with the summation running over all 128 dimensions.

3.8.2 Identification of k -nearest neighbors

For a given test sample x , the algorithm identified the $k = 3$ training samples with the smallest Euclidean distances specified by the $n_neighbors$ hyperparameter. This hyperparameter ($N_k(x)$) was determined by sorting the distances and selecting the top 3 samples.

3.8.3 Weighted voting with distance

The $weights = "distance"$ hyperparameter assigned votes to each neighbor inversely proportional to its distance enhancing the influence of closer matches. Mathematically, the weight w_i for the neighbor x_i was calculated as:

$$w_i = \frac{1}{d(x, x_i)^2 + \epsilon} \quad (7)$$

Where, ϵ is a small constant to avoid division by zero if $d(x, x_i) = 0$.

The predicted student name was then determined by a weighted majority vote given by:

$$y = \underset{c}{\operatorname{argmax}} \sum_{x_i \in N_k(x)} w_i \cdot I(y_i = c) \quad (8)$$

Where, c is a class label (i.e., the student's name), y_i is the label of its neighbor x_i , and I is an indicator function (1 is true and 0 is false).

3.9 Training of the multi-facet facial recognition model

The training process built the multi-facet facial recognition model by storing the training encodings and labels in memory. No explicit parameter optimization occurred during the training, as K-NN is instance-based and predominantly relied on the precomputed distances during prediction. The training process simply prepared the data structure for efficient nearest neighbor searches. Next, in order to enable deployment and reuse of the model, it was saved using the pickle library after the training process. This serialization technique stored the trained model instance, including the 128-dimensional facial encodings and corresponding labels, into a binary file, eliminating the need for retraining on each application. As a result of this, during prediction, a new facial encoding was compared against the training set, and the class with the student's encoding with the highest vote among the 3 nearest neighbors determined the student's identity. A 0.6

Table 2 Hyperparameters used for the multi-facet facial recognition model

Hyper-parameter	Value used	Description
n_neighbors	3	The n_neighbors value of 3 focused on the most similar encodings for the 3 students, which was important for distinguishing the students.
weights	distance	The distance weighting scheme assigned votes inversely proportional to the Euclidean distance, prioritizing closer facial encodings to improve accuracy for the conditions in varying lighting or head positions.
algorithm	auto	This selected the best search method from “ball_tree” or “kd_tree” based on the 128-dimensional facial encoding to handle real-time checks.
leaf_size	30	The leaf size of 30 provided a balanced memory usage and search speed, suitable for high-dimensional encoding, while ensuring quick classification of students in dynamic examination environments.
p	2	Specifying $p=2$ set the minkowski distance to Euclidean, which was well-suited for the 128-dimensional space of facial encodings.

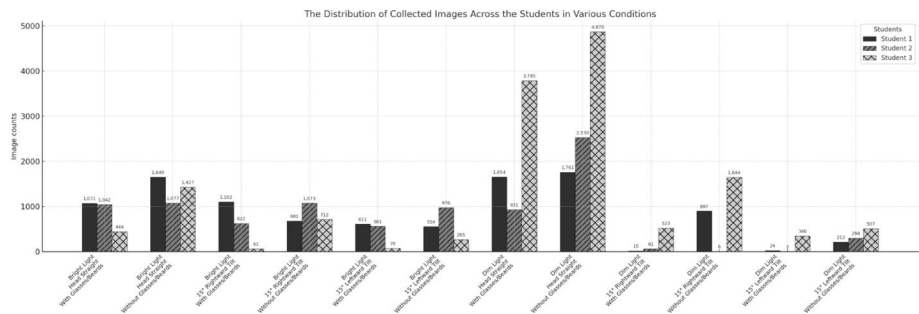


Fig. 5 Distribution of collected images across the students in bright and dim light conditions

threshold on the minimum distance was applied to flag unrecognized faces, aligning with the *unknown person* protocol (Table 2).

4 Discussions of results

4.1 Data collection results

To ensure the variability necessary for the multi-facet facial recognition model, the images were captured under a variety of conditions, namely:

- i. *Lighting variations*: Images were taken in both bright and dim light scenarios under different shades for real-world diversity.
- ii. *Head positions*: Three distinct head orientations were collected: straight, 15° leftward tilts, 15° rightward tilts to account for positional variability.
- iii. *Accessory and appearance variation*: Images were captured with and without glasses and with and without beards to reflect different facial appearances of the students.

A comprehensive summary of the dataset across the 3 students captured in bright and dim light scenarios is shown in Fig. 5.

4.2 Image resizing and gray scaling results

Figure 6 presents the student images following pre-processing. Images were then resized 400×400 to pixels to standardize dimensions for consistent feature extraction and computational efficiency. In addition, they were converted from BGR format to gray-scale, reducing processing overhead by eliminating redundant color data.

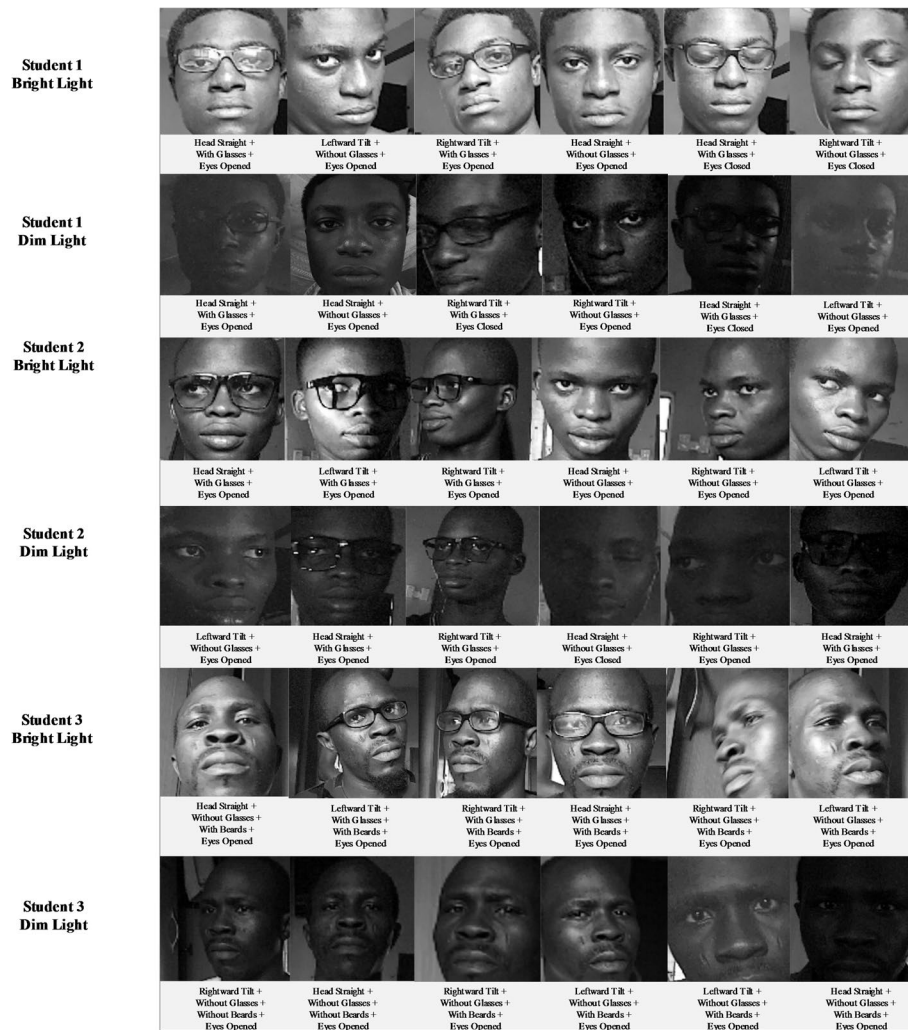


Fig. 6 Collection of the data after pre-processing

4.3 Data-preparation results

Figure 7, respectively, presents the dataset distribution in bright and dim light scenarios used for training the model, reflecting a curated selection process that reduced the initial collection to a total of 990 images by selecting the most representative samples across all variations for each of the three students. This ensured the inclusion of the best images to achieve a good performance while mitigating potential bias in the recognition process.

4.4 Feature extraction and encoding results

4.4.1 Face alignment

Figure 8 illustrates the results of the face alignment process, which involved identifying 68 landmark features for each student's image processed using dlib's ensemble of regression trees. These landmarks spanned unique facial regions of each student, including the eyes, eyelids, eyebrows, jawline, mouth, and nose, with additional points delineating the cheeks and chin to ensure precise alignment despite variations such as head tilts or the wearing of accessories. This alignment enabled consistency of facial orientations, which was necessary for subsequent encoding and recognition accuracy under various conditions, such as lighting changes or the presence of glasses or beards.

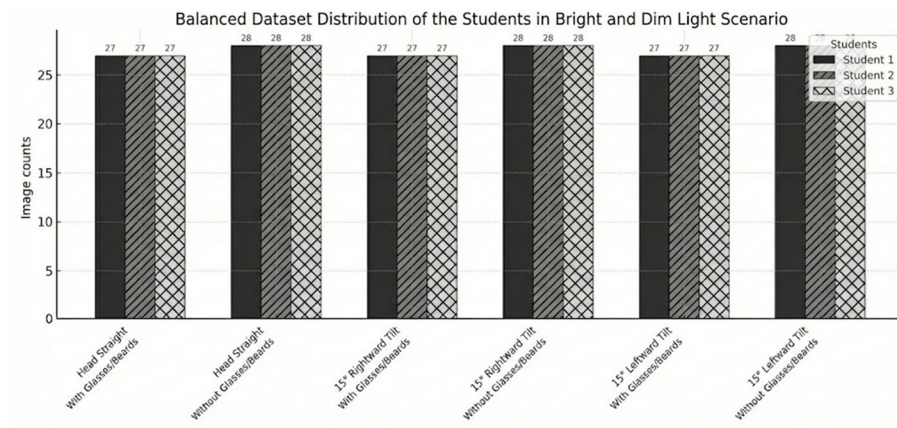


Fig. 7 Balanced dataset distribution of the students in bright and dim light scenario

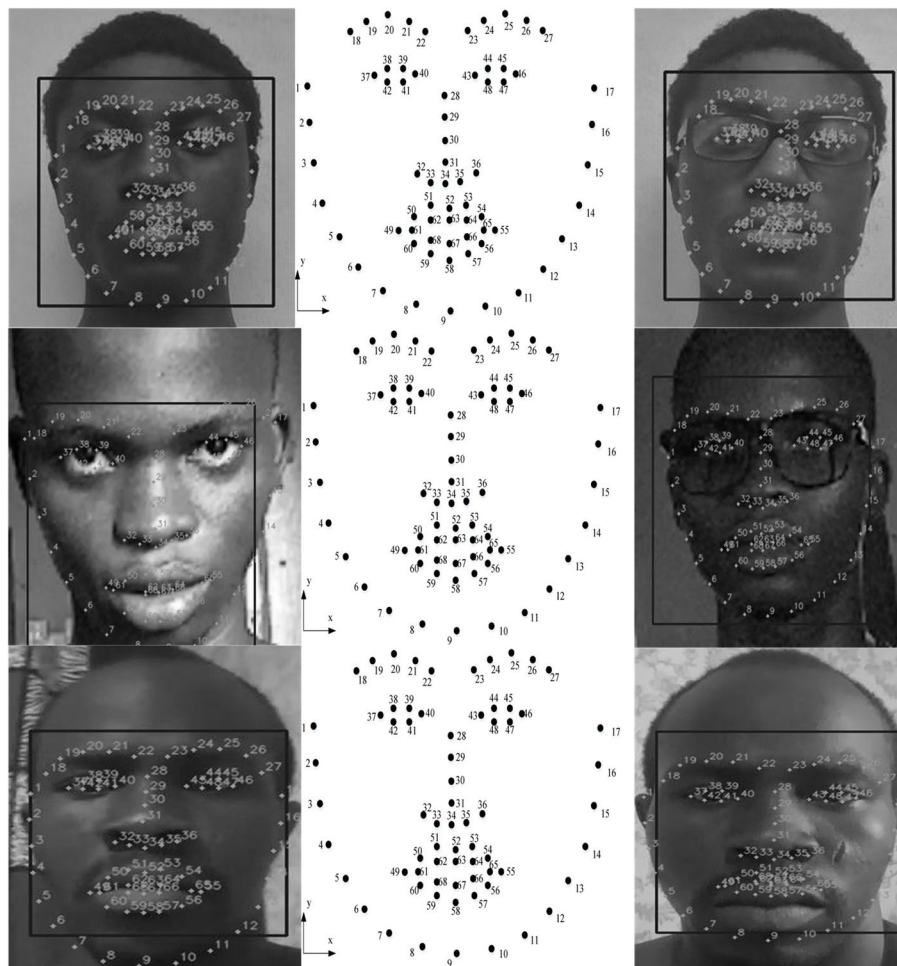
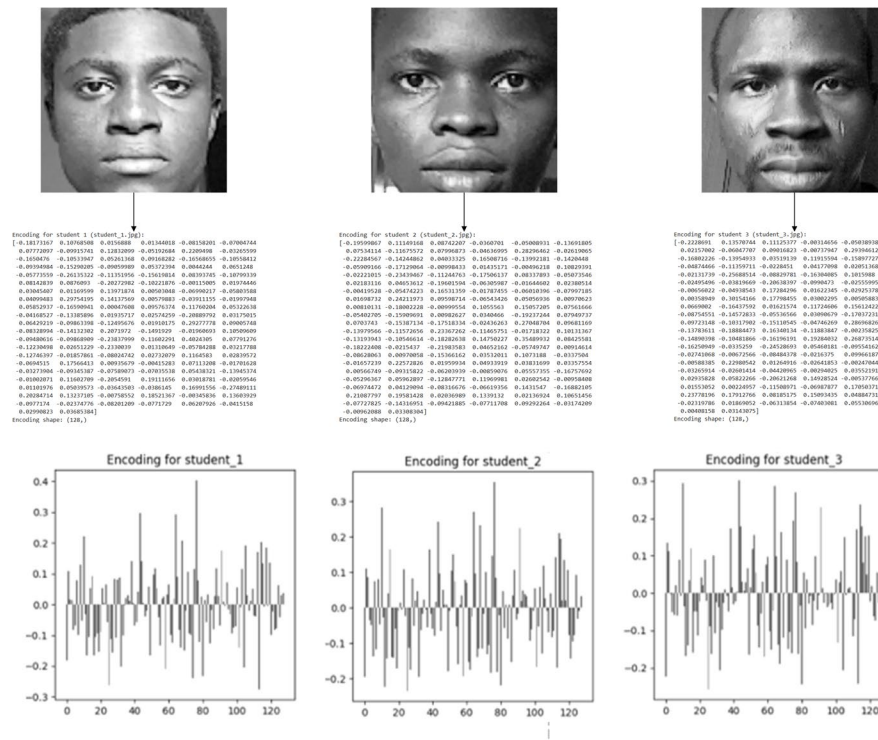


Fig. 8 The face alignment procedure

4.4.2 Face encoding

Figure 9 illustrates the results of encoding each aligned student's face into a 128-dimensional vector using the ResNet deep learning architecture. These encodings captured the distinctive facial features in a compact Euclidean space, enabling differentiation of

**Fig. 9** The face encoding procedure**Table 3** Performance metrics of the facial recognition model on images of each student

Student	Images processed	Images not processed	Precision	Recall	F1-score
Student 1	629	371	1.0000	1.0000	1.0000
Student 2	462	538	1.0000	0.9978	0.9989
Student 3	659	341	0.9984	1.0000	0.9992

students across variations, as depicted by the clustered vector representations in the figure. The three bar charts showed the 128-dimensional face encodings produced for each student. Each bar represented one component of the encoding vector; its height corresponded to the signed magnitude of the encoded feature. Positive and negative values did not map to single visible attributes but were simply numeric weights encoded by the network, forming a compact blueprint for each of the students' faces. Visually encoding patterns for the three students exhibited distinct shapes: some dimensions had relatively large magnitudes for one student but small or opposite-signed magnitudes for another. These consistent differences across multiple dimensions indicated that the model produced separable representations for the three identities, meaning the embeddings encoded identity-relevant information differently for each student, which is why the model could discriminate between them.

4.5 Performance evaluation

4.5.1 Detection stage analysis

The system's performance was evaluated in two distinct stages: face detection (Haar Cascade) and identity recognition (ResNet + KNN). Analyzing the detection stage first was important for understanding the operational boundaries of the system. As detailed in Tables 3 and 4, the face detection module successfully detected faces in 1750 out of 3000

Table 4 Performance metrics of the facial recognition model on video frames of each student

Student	Frames processed	Frames not processed	Precision	Recall	F1-score
Student 1	1215	329	0.9959	0.9934	0.9946
Student 2	492	207	0.9979	0.9593	0.9782
Student 3	2352	578	0.9903	0.9996	0.9949

static images (58.3% success rate) and 4059 out of 5,173 video frames (78.5% success rate).

The excluded instances (41.7% of images and 21.5% of video frames) represent “Failure to Detect” (FTD) events. These were attributed to detection conditions specifically extreme acute head angles or low-contrast illumination that exceeded the detection parameters of the Haar Classifier. While these detection failures do not produce False Acceptances (i.e. security breaches), they indicate that the standard Haar detector acts as bottleneck for raw data intake, filtering out suboptimal frames before they reach the recognition stage.

4.5.2 Recognition stage analysis for static images

The performance of the proposed recognition model was evaluated on the 1750 successfully detected images. This isolation ensured that the recognition metrics reflected the discriminative power of the ResNet embeddings and KNN classifier, independent of the detector.

As presented in Table 3, the model demonstrated accuracy on the processed set. Precision scores for Student 1 and 2 was 1.0000 indicating zero false positives. Similarly, the recall scores ranged from 0.9978 to 1.0000. These results confirmed that once a face was successfully localized, the generated 128-dimensional embeddings maintained high intra-class compactness and inter-class separability, even when the subject was presented with accessories (glasses/beards) or varying illumination.

To further analyze the classification performance, the confusion matrix presented in Fig. 9 illustrates the recognition breakdown for the 1750 processed images. As observed, all instances for Student 1 were correctly recognized with zero misclassifications. For student 2, a single false positive occurred where the model recognized “Student 3”. Similarly, Student 3 had one instance misclassified as “Student 2”. These minimal errors indicated that the model achieved a high level of inter class separability with minimal confusion between identities.

4.5.3 Recognition stage analysis for video data

The video evaluation further validated the model’s consistency in dynamic environments. Out of 4,059 processed frames, the system correctly classified 4030, achieving an overall recognition accuracy of 99.29%. As shown in Table 4, the F1-scores across all three students remained above 0.97, demonstrating that the embeddings were stable even when extracting features from motion blurred or continuously shifting video frames.

However, it is important to contextualize the detection rates reported in Sect. 3.5.1 for real-world deployment. In a single-shot verification scenario, a 41.7% detection failure rate (as seen in the static images) would result in a high user rejection rate, but the proposed system is designed for continuous video monitoring, processing approximately 30 frames per second.

In continuous operational mode, detection failures function as temporary latency rather than system denial. If the detector fails to capture a face in frame t due to an extreme angle, the temporal redundancy of the video feed allows for successful capture in frame $t + n$ as the subject naturally adjusts their pose. This is evidenced by the 0% False Omission Rate (FOR) observed during real-time testing, where every genuine student was eventually verified within the response window. Thus while the limitations of the Haar detector introduce a processing delay, they do not compromise the ultimate reliability of the attendance verification process.

The classification performance on video data is depicted in the confusion matrix heatmap in Fig. 10. The diagonal entries confirm that the vast majority of video frames were correctly classified. Misclassifications were present but limited: Student 2 was misclassified as Student 1 in 15 frames, and Student 3 was misclassified as Student 1 in 8 frames. This slight increase in confusion compared to static images is attributed to motion blur during frame capture, yet the overall accuracy remains at 99.29%.

4.5.4 Real-time performance evaluation

Real-time performance evaluation for the model was implemented across three distinct conditions: lighting variations, head positions, and appearance variations, focusing on two key metrics: False Omission Rate (FOR) and response time. The False Omission Rate (FOR) was assessed for each student across three distinct head positions: head straight, 15° rightward tilt, and 15° leftward tilt. These assessments were conducted under both bright and dim lighting, considering appearance variations such as the presence or absence of glasses and beards. A total of six genuine recognition attempts were concluded for each student across these scenarios, with no omissions recorded in any instances, resulting in a 0% False Omission Rate (FOR) for the model.

Figure 11, illustrates the recognition outcomes for each of the three students under these varied conditions and Tables 5, 6, 7, 8, 9 and 10 present the response time evaluation results for Students 1, 2, and 3 under varying lighting conditions (bright and dim light), head positions and use of accessories.

Mathematically, the average response time in each of these scenarios was calculated as follows:

$$\text{Response Time (RT)} = \frac{\sum_{i=1}^N t_i}{N} \quad (18)$$

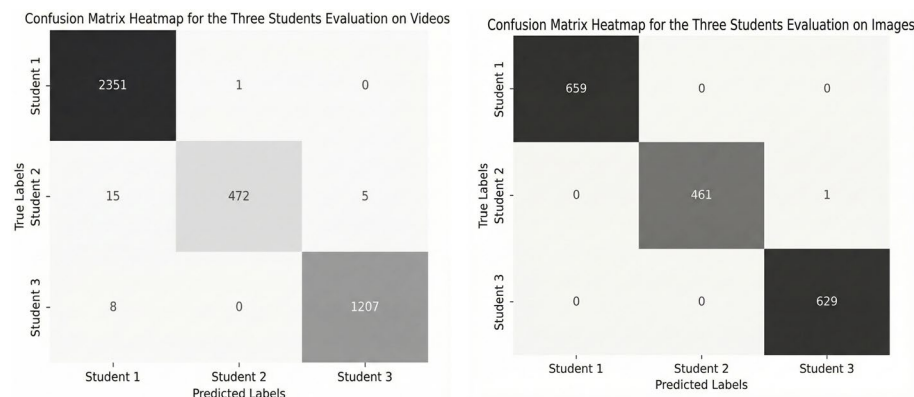


Fig. 10 Confusion matrix heatmap for the performance evaluation on images and videos

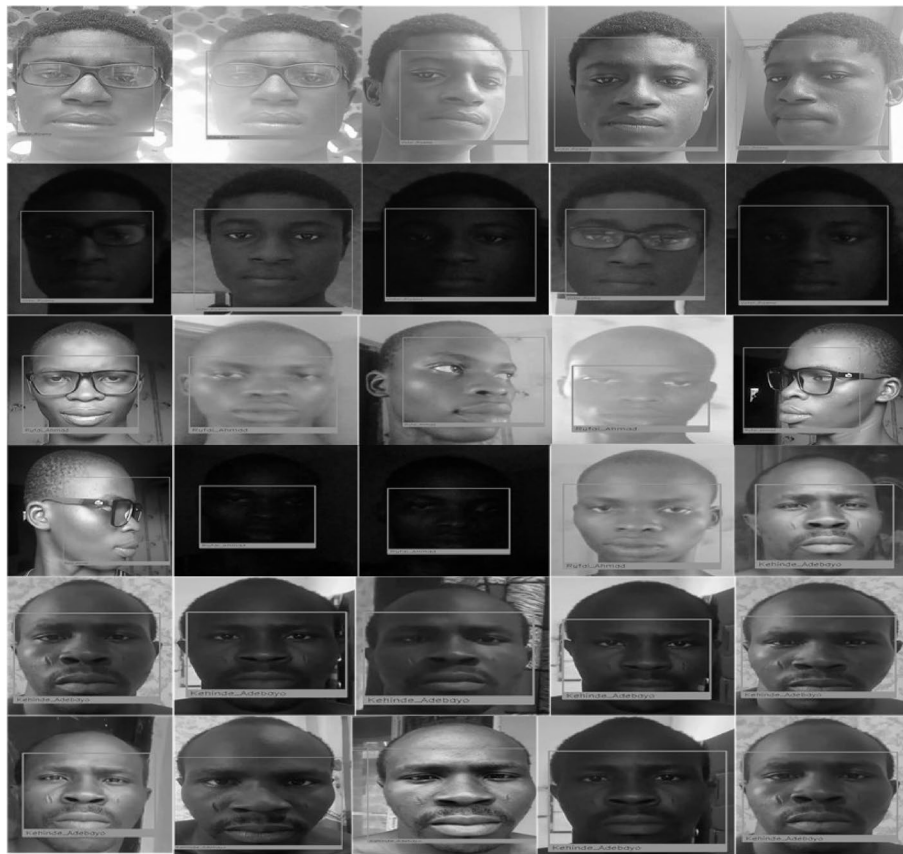


Fig. 11 The real-time recognition results for students 1–3 in the various conditions

Table 5 Response time results for student 1 in bright light

	Bright light					
	Head straight		15° rightward tilt		15° leftward tilt	
	With glasses/ beards	Without glasses/ beards	With glasses/ beards	Without glasses/ beards	With glasses/ beards	With- out glasses/ beards
Student 1						
1	3.61	3.54	3.34	3.54	4.26	3.79
2	3.52	3.44	4.69	5.56	4.32	4.51
3	4.12	4.61	4.92	4.66	3.72	3.84
4	3.79	3.68	4.38	4.82	4.16	4.27
5	3.93	3.77	4.53	4.74	4.08	4.12
Total processing time	18.97	19.04	21.86	23.32	20.54	20.53
Response time	3.79	3.81	4.37	4.66	4.11	4.11

Where, t_i was the time it took to process the i th test instance, N is the total number of test instances across the scenarios (e.g., images, video frames or live feed samples).

The model's performance in bright light conditions yielded an average response time of 4.02 s across the three students, indicating its ability for recognition in optimal illumination. In contrast, dim light scenarios extended the average time to 7.98 s, due to reduced feature visibility, yet this still underscored the model's adaptability in a low-light examination environment, demonstrating its practical utility where lighting variations were

Table 6 Response time results for student 1 in dim light

Student 1	Dim light					
	Head straight		15° rightward tilt		15° leftward tilt	
	With glasses/ beards	Without glasses/ beards	With glasses/ beards	Without glasses/ beards	With glasses/ beards	With- out glasses/ beards
1	9.51	3.91	8.07	3.93	3.66	5.47
2	6.48	3.42	5.01	4.50	3.51	4.49
3	7.83	3.76	6.74	4.21	3.82	4.95
4	8.15	4.10	7.12	4.39	4.01	5.23
5	7.24	3.85	6.43	4.07	3.74	4.88
Total processing time	39.21	19.04	33.476	21.10	18.74	25.02
Response time	7.84	3.81	6.69	4.22	3.75	5.00

Table 7 Response time results for student 2 in bright light

Student 2	Bright light					
	Head straight		15° rightward tilt		15° leftward tilt	
	With glasses/ beards	Without glasses/ beards	With glasses/ beards	Without glasses/ beards	With glasses/ beards	With- out glasses/ beards
1	5.66	4.20	5.32	3.47	3.78	3.78
2	3.97	4.00	3.55	3.57	3.79	3.02
3	3.67	4.30	6.21	3.57	3.92	3.88
4	3.84	3.90	5.64	3.39	3.49	3.39
5	3.90	3.99	8.74	3.80	3.68	3.14
Total processing time	20.98	20.39	29.46	17.80	18.66	17.21
Response time	4.20	4.08	5.90	3.56	3.73	3.44

Table 8 Response time results for student 2 in dim light

Student 2	Dim light					
	Head straight		15° rightward tilt		15° leftward tilt	
	With glasses/ beards	Without glasses/ beards	With glasses/ beards	Without glasses/ beards	With glasses/ beards	With- out glasses/ beards
1	4.28	4.40	13.12	13.45	11.94	11.94
2	4.51	4.56	15.89	16.35	13.92	14.20
3	4.06	4.32	14.78	15.12	13.35	13.64
4	4.37	4.25	14.23	14.77	12.85	13.11
5	4.19	4.34	13.96	14.58	12.46	12.88
Total processing time	21.41	21.87	71.98	74.27	64.52	65.77
Response time	4.28	4.37	14.40	14.85	12.90	13.15

common. When accessories such as glasses or beards were present, the average response time was 6.21 s, indicating additional computational demands to account for occluded facial features, compared to 5.79 s without them. This difference illustrates the model's resilience in handling real-world variations, effectively underscoring its ability for recognition of different student appearances during attendance activities. In head straight

Table 9 Response time results for student 3 in bright light

Student 3	Bright light					
	Head straight		15° rightward tilt		15° leftward tilt	
	With glasses/ beards	Without glasses/ beards	With glasses/ beards	Without glasses/ beards	With glasses/ beards	With- out glasses/ beards
1	3.12	3.97	3.81	3.70	3.64	3.71
2	3.67	3.44	3.63	3.49	3.69	3.72
3	3.84	3.44	5.92	3.76	3.96	3.91
4	3.58	3.73	3.88	3.65	3.77	3.82
5	3.47	3.61	3.79	3.68	3.73	3.85
Total processing time	17.68	18.19	21.03	18.28	18.79	19.01
Response time	3.54	3.64	4.21	3.66	3.76	3.80

Table 10 Response time results for student 3 in dim light

Student 3	Dim light					
	Head straight		15° rightward tilt		15° leftward tilt	
	With glasses/ beards	Without glasses/ beards	With glasses/ beards	Without glasses/ beards	With glasses/ beards	With- out glasses/ beards
1	4.59	4.37	5.12	3.69	14.25	14.62
2	4.04	4.10	5.47	5.79	14.68	15.00
3	4.36	4.22	5.63	4.96	14.81	14.74
4	4.41	4.18	5.29	5.11	14.53	14.87
5	4.33	4.25	5.51	5.03	14.76	14.90
Total processing time	21.73	21.12	27.02	24.58	73.03	74.13
Response time	4.35	4.22	5.40	4.92	14.61	14.83

positions, the model achieved an average of 4.33 s, showing a strong baseline accuracy, while 15-degree leftward tilt (LWT) prolonged it to 7.27 s and 15-degree rightward tilt (RWT) to 6.40 s, owing to asymmetrical feature distortions. These emphasized the model's capability to manage minor head position deviations, reinforcing its relevance for dynamic examination scenarios where perfect alignment may not always be feasible. Overall, the model's average response time of 6.00 s across all tested variations signified a balanced and reliable system.

However, regarding the practical deployment of this model in large scale examination halls, the reported response times specifically the 7.98 s average in dim lighting warrant operational consideration. While sub-second processing is typical for low stakes application, high stakes applications like examination verification prioritizes False Acceptance Rate (FAR) minimization over throughput speed. In this context, a verification latency of 4–8 s is deemed acceptable to ensure the integrity of the candidate screening process. Nevertheless, for halls with simultaneous entry of hundreds of candidates, a single verification unit operating at this speed could create ingress bottlenecks. To mitigate this in large cohorts, the system's modular design allows for the deployment of multiple parallel verification nodes at the point of entry. This approach maintains the rigorous security checks of the model while accommodating large candidate volumes within the allotted entry window.

4.6 Comparative analysis of lightweight classifiers

To validate the choice of the KNN classifier, a comparative study was conducted against SVM and MLP classifiers using the same 128-dimensional embeddings. The evaluation was performed on both the static image dataset of 3000 images and the collected video dataset.

Table 7 presents the performance benchmarks. On the static image dataset, the Linear SVM achieved an accuracy of 99.94%, while the MLP achieved 99.66%, both performing comparably to the proposed KNN model.

Classifier	Image accuracy	Video accuracy	Video precision	Video recall	Video F1-score
KNN	99.94%	99.29%	99.47%	98.41%	98.92%
SVM	99.94%	99.12%	99.37%	97.49%	98.42%
MLP (64-Neurons)	99.66%	99.25%	99.37%	98.12%	99.31%

The results indicate that the system's high performance is primarily driven by the discriminative power of the ResNet embeddings rather than the specific classification head. However, the slightly higher complexity of the MLP did not yield a statistically significant performance gain over KNN or SVM. Consequently, KNN remains an optional choice for this application due to its negligible training time and support of incremental student enrollment without retraining.

5 Conclusion

This study successfully developed a multi-facet facial recognition model to authorize attendance during high-stakes examinations. By integrating Haar Cascade detection with ResNet-based feature encoding and KNN classification, the system generated highly discriminative 128-dimensional vectors capable of verifying identities despite variations in lighting, head orientation, and facial accessories. Evaluation across static images, video feeds, and real-time scenarios confirmed the model's robustness, achieving an overall high recognition accuracy and a False Omission Rate (OR) of 0% during live deployment.

However, practical limitations were identified regarding operational efficiency and generalizability. While the recognition module proved highly resilient, the face detection stage exhibited sensitivity to extreme head poses and low-contrast illumination, resulting in "Failure to Detect" events that necessitate user cooperation during scanning. In addition, the reliance on a controlled cohort limits the assessment of inter-class separability; scaling to larger populations typically shrinks the Euclidean margins between identities, implying that the reported accuracy serves as a robust baseline that may require threshold optimization for mass deployment. Also, the average response time of approximately 6.00 s, while sufficient for security-focused verification, implies that large-scale deployment requires a distributed architecture to prevent ingress bottlenecks in crowded examination halls.

Based on the study's findings, future work should address scalability and robustness challenges with specific attention given to distinguishing look-alike faces (e.g. identical twins) by creating specialized impostor datasets. Despite the current model's effective encoding of unique facial features for each individual into 128-dimensional vectors which provided strong discriminative power, more comprehensive testing with look-alike scenarios will further strengthen its robustness. Furthermore, incorporating hybrid local-pattern-based facial analysis would enhance system scalability. By integrating

computationally efficient descriptors like RTLBP and MVM-LBP with deep learning embeddings, the system would reduce computational burden of high-dimensional feature matching, enabling faster retrieval in large-scale cohorts. Additional improvements include integrating liveness detection methods into the model development to prevent spoofing and expanding the dataset with extreme lighting, wider head tilts, and diverse accessories. Ultimately, transitioning to a hybrid approach while combining these local descriptors with deep learning architectures would offer the optimal balance of accuracy, real-time efficiency, and robustness for institutional deployment.

Author contributions

Kufre Esenowo Jack contributed to the conceptualization, methodology, supervision, overall administration of this research, and final approval of the manuscript for submission. Victor Samuel Rizama conducted the literature review, data collection, data preprocessing, model development and training, model testing and evaluation, and prepared the initial draft of the manuscript. Kehinde Rahmon Adebayo contributed to data collection, literature review, result analysis, and algorithm implementation. James Garba Ambafi supported manuscript editing, data preprocessing, validation, and preparation of figures and tables. Rufai Ahmad Olayemi provided literature review, theoretical analysis, and interpretation of results. John Oluwatobi Olaoye performed the analysis, performance evaluation, and contributed to manuscript editing. Damilola Olanipekun Tunbosun offered technical support and revised the manuscript. Adeshina Victor Olawuyi contributed to manuscript review and editing.

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Data availability

The dataset generated during this study is not publicly available because it contains facial images of some of the participating researchers, which raises privacy and confidentiality considerations. However, as it is demanded by Journal Editor, the processed dataset supporting the findings of this study is available at the link herein for your assessment Facial Recognition Dataset.

Declarations

Ethics approval and consent to participate

Ethical approval for this study was obtained from the Ethics Committee of the Federal University of Technology, Minna, Niger State, Nigeria. All procedures involving human participants for the collection of photographic data were conducted in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. Informed consent was obtained from all individual participants included in the study. Participants were fully informed regarding the purpose of the data collection and how their facial images would be utilized for training and evaluating the facial recognition model.

Consent for publication

The authors affirm that human research participants provided informed consent for the publication of the recognizable facial images included in the figures of this manuscript.

Competing interests

The authors declare no competing interests.

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